A Collaborative RC QC-LDPC Code Construction Scheme Using Both Extension and Splitting

Huang-Chang Lee, *Member, IEEE*, Jyun-Ching Wang, Yi-Tsun Huang, and Yeong-Luh Ueng, *Senior Member, IEEE*

***Abstract*— This letter presents a construction flow for rate-compatible LDPC (RC-LDPC) codes, collaboratively using both extension and splitting schemes. In addition to the decoding threshold used for selecting the degree distribution, a new metric, known as the Average ACE (A^2CE) value, is introduced in order to estimate the Tanner graph connectivity. The proposed collaborative scheme provides more flexibility in the designs of the degree distribution and graph connectivity, both of which are critical for RC LDPC codes that have short and/or moderate lengths. The designed LDPC codes achieve a comparable or better error rate performance compared to existing codes.**

***Index Terms*— Channel coding, low-density parity-check code, rate-compatible, extension, splitting, ACE, RC-LDPC codes.**

I. Introduction

WHEN constructing rate-compatible low-density parity-check (RC-LDPC) codes, we can rely on schemes such as puncturing [1], extension [2]–[4], [18], splitting [5], [6], and shortening. In addition, a number of hybrid schemes are based on an intermediate-rate code, where high-rate codes are constructed using puncturing, and low-rate codes are constructed using either shortening or extension [7], [13]–[16]. In this letter, we propose using a collaborative approach that allows each child code to be selected from two different schemes such that a more flexible design over a wide range of code rates can be achieved. Codes constructed from both extension and splitting schemes are then used as candidates for the low-rate child code.

In the proposed collaborative scheme, the decoding threshold in terms of E_b/N_0, which is determined by the degree distribution, must be taken into consideration. A code with a lower decoding threshold is preferred since its decoding process converges at a lower signal-to-noise ratio (SNR). Since we focus on codes that are of a short to moderate length, the connectivity of the Tanner graph also has a great impact on the error rate performance [9]. To estimate the connectivity, the degree of the approximate cycle extrinsic (ACE) message, which is the maximum possible degree of an extrinsic message for a cycle, and the ACE spectrum, which provides the minimum ACE values for cycles that have different lengths, can be used [10]. This metric has been applied to the construction of both the extension codes [3] and the puncture codes [12].

For the extension scheme, the same number of new rows and new columns are added to the parity-check matrix (PCM) for the low-rate child code. Since the PCM for the high-rate parent code is embedded in the low-rate child code, the extension code usually has a better connectivity, even if the selection of the degree distributions is limited. In contrast, when the splitting scheme is used, a single row in the high-rate PCM is split into two rows, and a weight-2 column is added to connect the two new rows in the low-rate PCM. It can be expected that the number of additional connections is limited in the splitting codes, but they usually contain fewer short cycles and tend to enlarge the girth of the Tanner graph.

Compared to splitting codes, extension codes usually contain more short cycles, but each of these cycles has a better external connection and, hence, has a larger ACE value. In contrast, splitting codes are usually more sparse, with fewer external connections to each cycle, and also have lower ACE values. Since the ACE spectrum only provides the minimum ACE values for cycles of different lengths, an indicator, known as the average ACE (A^2CE) value, is introduced in this work to indicate the selection of codes constructed using either the extension or the splitting schemes. The A^2CE value is defined as the ratio between the total number of ACE values for all the shortest cycles and the total number of the shortest cycles in the Tanner graph.

Extension codes and splitting codes have different characteristics in degree distribution and connectivity. In the proposed collaborative scheme, codes constructed using both schemes are considered as candidates for the child code and hence additional varieties of degree distribution and connectivity as well as error-rate performance are possible. The designed RC QC-LDPC codes from rate-$4/5$ to $1/3$ show a comparable or better error rate performance when compared to those codes constructed based on pure extension [3], [4] and pure splitting [5] schemes. From the comparison between the proposed codes and the existing codes that include the 5G codes, the proposed collaborative construction scheme using both splitting and extension techniques has an advantage over other hybrid approaches.

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Huang-Chang Lee is with the Department of Electrical Engineering, Chang Gung University, Taoyuan 33302, Taiwan, and also with the Chang Gung Memorial Hospital, Chang Gung University, Taoyuan 33302, Taiwan (e-mail: akula.lee@gmail.com).

Jyun-Ching Wang was with the Department of Electrical Engineering, National Tsing Hua University, Hsinchu 300, Taiwan. He is now with HTC Corporation, New Taipei City 231, Taiwan.

Yi-Tsun Huang was with the Department of Electrical Engineering, National Tsing Hua University, Hsinchu 300, Taiwan. He is now with NXP Semiconductors Taiwan Limited, Kaohsiung 811, Taiwan.

Yeong-Luh Ueng is with Department of Electrical Engineering and Institute of Communications Engineering, National Tsing Hua University, Hsinchu 300, Taiwan (e-mail: ylueng@ee.nthu.edu.tw).

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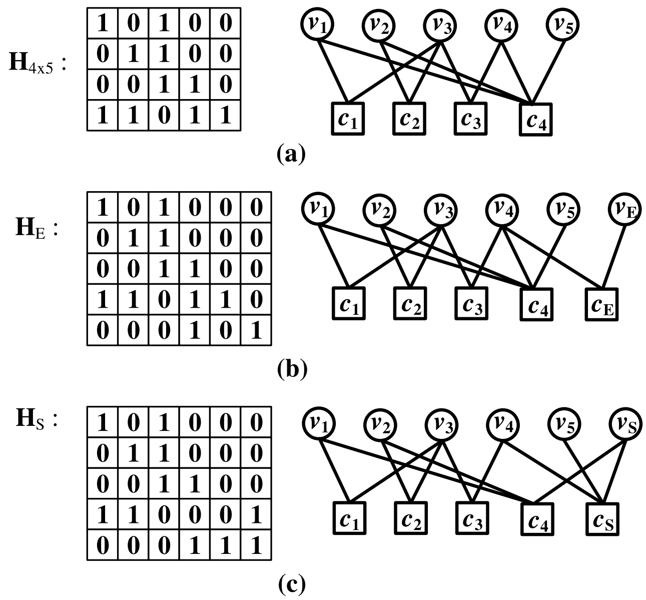

Fig. 1. The extension and splitting codes for the evaluation of both the minimum ACE and the A^2CE values.

The remainder of this letter is organized as follows. The indicator A^2CE is presented in Section II. In Section III, A^2CE is adopted in the design flow for the proposed code. The code families are realized and their performance is evaluated in Section IV. Finally, Section V concludes this work.

II. ACE Property for LDPC Code Construction

In general, a well-constructed LDPC code should have a small number of short cycles in its Tanner graph, and each cycle should have a large number of external connections. In [10], the ACE value is used to serve as an estimate of the external connections.

A. ACE Evaluation for Extension and Splitting

The LDPC codes constructed in [3] and [10] only refer to the minimum ACE value for the shortest cycles. This value is denoted as η_{ACE}. Since the minimum ACE value may not be an appropriate selection parameter for the candidate child codes, we use the average ACE value in the proposed collaborative scheme, which is the ratio between the total number of ACE values for all the shortest cycles and the total number of shortest cycles in the Tanner graph. The necessity of the proposed A^2CE can be demonstrated using the example shown in Fig. 1. Suppose that the PCM for the code is $H_{4\times5}$, where the corresponding Tanner graph is shown in Fig. 1 (a). For simplicity, only the two 6-cycles are considered, which are $S_1 = \{v_1, c_1, v_3, c_2, v_2, c_4\}$ and $S_2 = \{v_2, c_2, v_3, c_3, v_4, c_4\}$. For all variable nodes in both cycles, v_3 is the only variable node that includes a single external connection, then the ACE values for both S_1 and S_2 are equal to 1, and the minimum ACE value for the 6-cycles is 1.

When the extension scheme is applied to the construction of the RC-LDPC code, an additional row and an additional weight-1 column are added. The resultant PCM H_E and the corresponding Tanner graph are shown in Fig. 1 (b), where a new variable node v_E and a new check node c_E are added. The external connections for cycle S_1 remain the same after the low rate code that includes PCM H_E is constructed, and the additional check node c_E is connected to variable node v_4 in S_2. The ACE value for S_2 is subsequently increased from 1 to 2. However, the increase in the ACE value for S_2 cannot be observed via the minimum ACE value.

When the splitting process is applied to matrix $H_{4\times5}$, the 4th row of $H_{4\times5}$ is split, and an additional weight-2 column is added. The resultant PCM H_S and the corresponding Tanner graph are shown in Fig. 1 (c). The 4th row of $H_{4\times5}$ is split into the 4th and the 5th rows in H_S, and the two additional edges are used to connect the new check node c_S corresponding to the additional column in H_S. Cycle S_2 has been broken, but the external connections for cycle S_1 remain the same as those of $H_{4\times5}$. Now the splitting code only contains one 6-cycle, S_1, whose ACE value is 1, then the splitting code has a minimum ACE value of 1, which is equal to that of the extension code.

B. Proposed Average ACE Value

The extension code shown in Fig. 1 (b) contains two 6-cycles, where the ACE value for S_1 is 1, and is 2 for S_2. In contrast, the splitting code shown in Fig. 1 (c) contains only a single 6-cycle, i.e., S_1, whose ACE value is 1. Although the extension code has more 6-cycles, the external connections for some 6-cycles are better than those of the splitting code. It is obvious that the minimum ACE value for these two codes is the same and, hence, the difference in the connectivity characteristics between them cannot be observed through the minimum ACE values.

According to our experiments, there can be a large difference between the number of cycles in both the extension codes and the splitting codes. In order to provide more complete information for the connectivity of the candidate child codes, both the number of cycles and the summation of the ACE values should be evaluated in addition to the minimum ACE value. For simplicity, only the ACE values for the shortest cycles are considered in this work. Suppose that 4-cycles are avoided in all candidate codes, and let the ACE value for the j-th 6-cycle be denoted as $ACE_6(j)$, the A^2CE value is defined as:

$$A^2CE = \left(\sum_{j=1}^{N_{C6}} ACE_6(j) \right) / N_{C6}, \tag{1}$$

where N_{C6} is the total number of 6-cycles. The A^2CE value for the extension code and the splitting code are denoted as A^2CE_E and A^2CE_S, respectively.

For the example shown in Fig. 1 the original code defined using $H_{4\times5}$ contains cycle S_1, which has an ACE value equal to 1, and cycle S_2 that has an ACE value that is also equal to 1, meaning that its A^2CE value is equal to $(1+1)/2 = 1$. The extension code defined using H_E contains cycle S_1, which has an ACE value of 1, and cycle S_2 that has an ACE value of 2, meaning that $A^2CE_E = (1+2)/2 = 1.5$. The splitting code defined using H_S only contains cycle S_1, which has an

ACE value of 1, so $A^2CE_S = 1$. Since $A^2CE_E > A^2CE_S = A^2CE$, we can say that, from the view point of the average ACE value, the connectivity for the extension is improved compared to the high-rate parent code, but the splitting code has an equal connectivity.

III. Proposed Collaborative Scheme

Let the rate of the parent code be denoted as R_i. Multiple candidates for the child code that has a rate R_{i+1}, where $R_{i+1} < R_i$, are generated using both the extension and splitting schemes. Among all candidate extension codes, the one that has the lowest decoding threshold T_E is denoted as C_E, and the value of A^2CE_E is evaluated. Similarly, code C_S has the lowest decoding threshold T_S among all the candidate splitting codes, and the A^2CE_S value is also evaluated. Initially, the selection of the construction scheme depends on the decoding threshold. If the difference between T_E and T_S is larger than a pre-determined positive value ΔT, the candidate code that has the lower decoding threshold can be directly selected. Since an appropriate value of ΔT varies with the code length, the size of the base matrix, and the value of the lifting factor, $\cdots$, etc., it is difficult to obtain an analytical value for ΔT, and, hence, an experiential-based value is used. Fortunately, the performance of the designed codes is not sensitive to the value of ΔT based on our experience. We also define the difference between A^2CE_E and A^2CE_S as $\Delta A^2CE_{SE} = A^2CE_S - A^2CE_E$. When the difference between the decoding threshold for C_E and C_S is smaller than ΔT, the selection depends on both the ΔA^2CE_{SE} value and the scheme used to construct the parent code. The details of the construction flow are presented in Algorithm 1.

Algorithm 1 Proposed RC-LDPC Code Construction Algorithm

1: Identify the highest-rate parent code and its base matrix, $i = 0$.
2: Parent code C_i has rate R_i and A^2CE_i.
3: Child code C_{i+1} has rate R_{i+1} and A^2CE_{i+1}, $R_{i+1}<R_i$.
4: Determine the extension code C_E for rate R_{i+1}.
5: C_E has T_E and A^2CE_E.
6: Determine the splitting code C_S for rate R_{i+1}.
7: C_S has T_S and A^2CE_S.
8: **if** $T_E - T_S > \Delta T$: C_{i+1}=C_S, $A^2CE_{i+1} = A^2CE_S$.
9: **elseif** $T_S - T_E > \Delta T$: C_{i+1}=C_E, $A^2CE_{i+1} = A^2CE_E$
10:**elseif** $|T_E - T_S| < \Delta T$
11: **if** C_i is a splitting code: C_{i+1}=C_E, A^2CE_{i+1}=A^2CE_E
12: **elseif** A^2CE_S>A^2CE_i: C_{i+1}=C_S, A^2CE_{i+1}=A^2CE_S
13: **else**: C_{i+1}=C_E, $A^2CE_{i+1} = A^2CE_E$
14: **end if**
15: **end if**
16: C_i=C_{i+1}, R_i=R_{i+1}, $A^2CE_i = A^2CE_{i+1}$, $i = i + 1$.
17: **if** R_i is not the lowest rate: go to Step **2**.
18: Terminate.

Algorithm 1 is used to construct RC QC-LDPC codes. From Steps 4 to 7, multi-edge-based matrices that have the lowest decoding threshold T_E and T_S are respectively determined for extension code C_E and splitting code C_S using the protograph extrinsic information transfer (P-EXIT) chart [8]. These multi-edge-based matrices are lifted to single-edge base matrices. Each of the non-zero elements in the single-edge-based matrix is then lifted to a z-by-z circulant permutation matrix, where z is the lifting factor, and the permutation indices are determined by using the RS-LDPC codes presented in [11]. This means that the single-edge-based matrix is used as a mask and is applied to the RS-LDPC codes. Since RS-LDPC codes avoid all 4-cycles, the resultant QC RC-LDPC codes do not contain any 4-cycles. The A^2CE_E and A^2CE_S values are evaluated according to the PCMs for C_E and C_S, respectively.

TABLE I
Coding Scheme at Rate-$\frac{8}{11}$

Rate	C_X	T_X(dB)	N_{C6}	$\sum_{j=1}^{N_{C6}} ACE_6(j)$	A^2CE_X	ΔA^2CE_{SE}
4/5	C_i	2.47	403	2373	5.88	
8/11	C_E	2.01	562	4707	8.375	
	C_S	2.31	227	1385	6.1	-2.275

Since more edges are added to the child code constructed using the extension scheme, the connectivity for the child code is usually better than for its parent code, although the number of short cycles is usually increased. In contrast, a candidate code constructed using the splitting scheme usually has fewer short cycles compared to its parent code. In order to prevent the number of short cycles from increasing too rapidly, we prefer to select child codes constructed from splitting and extension schemes alternatively, and this approach can be observed in the flow from Steps 11 to 13 in Algorithm 1. However, the selection of the splitting candidate is more strict, where Step 12 is used to ensure that its A^2CE value is larger than that of its parent code. This step is used to ensure that we are able to continue to increase the A^2CE values during the entire code construction process such that the connectivity can be guaranteed.

Table I shows an example for $R_i = 4/5$ and $R_{i+1} = 8/11$. Candidates C_E and C_S are prepared, and the decoding threshold for C_E is much better, i.e., $T_E = 2.01 < T_S = 2.31$. According to Step 9 in Algorithm 1, C_E is directly selected as the rate $R_{i+1} = 8/11$ code.

The example in Table II is used to show that the candidates C_E and C_S are selected alternatively by default. Candidates C_E and C_S have a similar value for the decoding threshold, i.e., $|T_E - T_S| = 0.05 < \Delta T = 0.25$, and C_E has a better average ACE value. Consider that the parent code that has a rate R_i is constructed using the extension scheme, then C_S is preferred as the child code where the rate is R_{i+1}, but the improvement in connectivity must be evaluated. The selection between C_E and C_S depends on the values of A^2CE_i and A^2CE_S. Since $A^2CE_S = 11.556 > A^2CE_i = 11.07$, it shows that C_S is still able to improve the connectivity, so C_S is selected as the rate $R_{i+1} = 8/13$ code, even though it has a slightly higher decoding threshold and a lower average ACE value compared to C_E. Using such a construction process, the number of shortest cycles, i.e., N_{C6}, for the selected rate-8/13 code will be reduced, which will be helpful for the construction of the following lower-rate codes.

TABLE II

CODING SCHEME AT RATE-$\frac{8}{13}$

Rate	C_X	T_X(dB)	N_{C6}	$\sum_{j=1}^{N_{C6}} ACE_6(j)$	A^2CE_X	ΔA^2CE_{SE}
2/3	$C_i = C_E$	1.57	752	8326	11.07	
8/13	C_E	1.14	1012	14135	13.967	
	C_S	1.19	347	4010	11.556	+0.486

TABLE III

CODING SCHEME FROM RATE-$\frac{4}{5}$ TO $\frac{1}{3}$

Rate	C_X	T_X(dB)	N_{C6}	$\sum_{j=1}^{N_{C6}} ACE_6(j)$	A^2CE_X
$4/5$		2.386	403	2373	5.88
$8/11$	C_E	2.01	562	4707	8.375
$2/3$	C_E	1.57	752	8326	11.07
$8/13$	C_S	1.19	347	4010	11.556
$4/7$	C_E	1.05	467	6469	13.852
$8/15$	C_S	0.65	162	2667	13.89
$1/2$	C_E	0.57	255	4007	15.713
$8/17$	C_S	0.41	206	3333	16.179
$4/9$	C_E	0.32	246	4474	18.186
$8/19$	C_E	0.21	291	5933	20.388
$2/5$	C_S	0.11	222	4457	20.765
$8/21$	C_E	0.01	261	5651	21.65
$4/11$	C_E	-0.1	329	7995	24.3
$8/23$	C_S	-0.19	253	6164	24.363
$1/3$	C_E	-0.25	359	9518	26.512

In summary, when $T_E < T_S - \Delta T$, candidate C_E for the child code can be directly selected, as, in general, C_E is able to improve the connectivity. However, when $T_S \le T_E - \Delta T$, the increase in the A^2CE value must be evaluated before candidate C_S can be selected. The goal of the proposed construction flow is to design a sequence of RC-LDPC codes that have steadily increasing A^2CE values and a lowering decoding threshold.

IV. PERFORMANCE EVALUATIONS

The RC-LDPC codes that have rates ranging from 4/5 to 1/3 have been constructed following Algorithm 1, where ΔT is set to 0.25 dB. The rate-4/5 AR4JA code described in [4] is used as the code that has the highest rate. The 4-cycles are removed from codes of all rates, and the A^2CE value is evaluated based on the 6-cycles only. The parameters for the resultant codes are shown in Table III. When a code is constructed using the extension scheme, the number of short cycles increases, but more external connections are also appended, meaning that the A^2CE value rapidly increases. However, for some rates, a code constructed using the splitting scheme enjoys a better decoding threshold compared to the extension candidate. If the A^2CE value can be steadily improved, then the splitting code can be considered as a candidate.

A. Comparison With Pure Extension and Splitting Constructions

The RC-LDPC codes have been realized based on an information length of 960 bits, i.e., $k = 960$, and the error-rate performance is evaluated using BPSK modulation in the AWGN channel, where the Sum-Product algorithm (SPA) is used. The rate-2/3, 1/2, and 1/3 codes are evaluated, and a comparison in the frame-error rate (FER) is presented in Fig. 2.

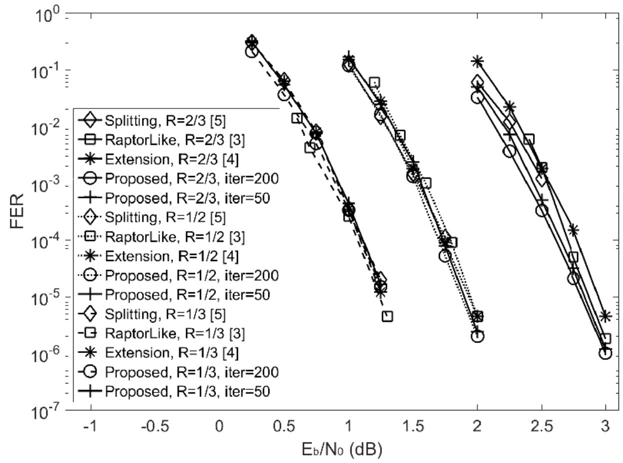

Fig. 2. FER for different rate-$\frac{2}{3}$, $\frac{1}{2}$, and $\frac{1}{3}$ codes, where R denotes code rate.

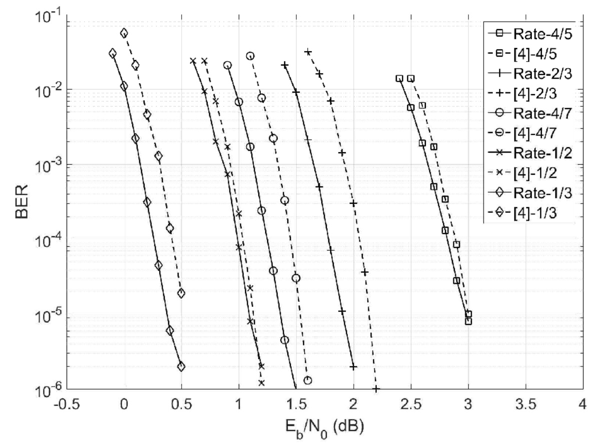

Fig. 3. BER for the code families that have an information length of 4106 bits constructed using the proposed method and those presented in [4].

The results for the pure extension codes [3], [4] and the pure splitting codes [5] are reported based on 200 and 50 iterations, respectively. The information lengths for the codes presented in [3], [4], and [5] are 1032, 1024, and 1056 bits, respectively. The proposed codes achieve the shortest information length, but are still able to achieve a comparable error-rate performance in almost all cases.

In [4], the decoding thresholds for those codes constructed using a pure extension scheme are listed. It can be observed that, in many cases, codes constructed using the pure extension scheme provide a better decoding threshold, including for rates $2/3$, $1/2$, and $1/3$. However, this does not match the performance shown in Fig. 2, where the codes constructed using Algorithm 1 provide a better FER performance for all three rates. It can also be seen from Fig. 3 that the error-rate performance for $k = 4106$ achieved by a family of RC-LDPC codes constructed using the proposed method is better than those presented in [4], where the same base matrix is used for the highest-rate codes in the two constructions. This implies that, for short to moderate codes, the error rate performance is not merely dominated by the decoding threshold, and Algorithm 1 including A^2CE metric can be

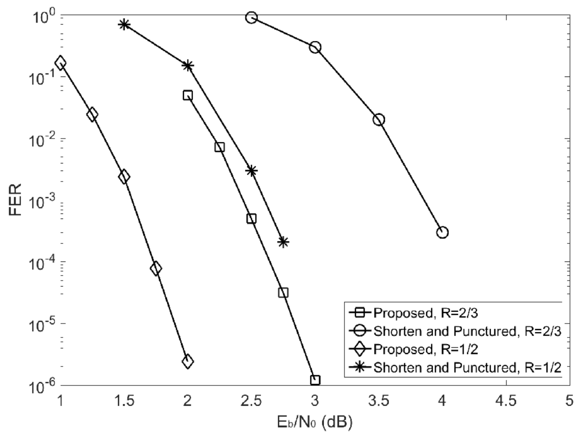

Fig. 4. FER for the proposed codes and the codes presented in [15], where k = 960 is considered.

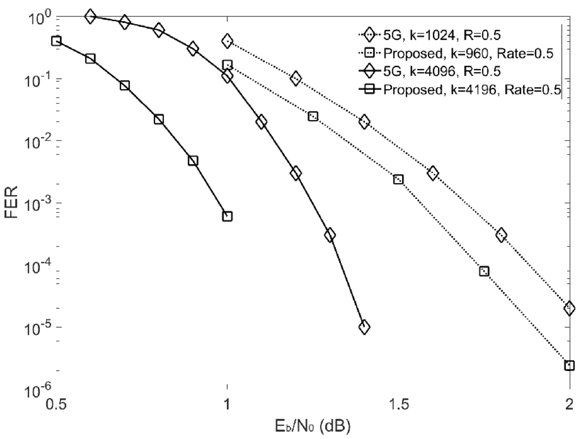

Fig. 5. FER for the proposed codes and the 5G codes presented in [17], where both k = 960 and k = 4160 are considered.

used. Although the results shown in this letter relate to short to moderate codes, the proposed collaborative scheme can be used to construct long codes, where the weighting between the threshold and the connectivity and, hence, the code selection parameters must be reconsidered in order to design suitable codes.

B. Comparisons to Other Hybrid Constructions

In [15] and [16], hybrid RC-LDPC code construction schemes using both shortening and puncturing are presented. In Fig. 4, the proposed codes are compared to the RC-LDPC codes constructed using the scheme presented in [15]. It can be observed that the codes constructed using the proposed collaborative scheme achieve a better performance than the codes presented in [15].

In Fig. 5, the proposed codes are compared to the 5G standard codes, which are also QC-LDPC codes. The 5G standard codes are basically constructed using extension, while the first two block columns are punctured. The FER performance attained by the 5G standard codes is referred to [17]. It can be seen that for both $k = 960$ and $k = 4160$, the proposed rate-1/2 codes provide a better FER performance.

V. CONCLUSION

We have presented a construction flow for RC-LDPC codes based on a collaboration between the extension and the splitting schemes. This flow provides a more flexible design for RC-LDPC codes over a wide range of code rates. In addition to the decoding threshold, the A^2CE value is introduced as an additional metric in the selection of child codes. The codes constructed using the proposed method have a comparable/better performance to/than existing RC-LDPC codes constructed using other approaches.

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